Emotion Classification of Chinese Weibo Posts

Classical Machine Learning with a Leakage-Free Evaluation & Interactive Demo

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1 Problem Statement

Inspired by Bai, He & Yuan (2021), *Sentiment Analysis of Chinese Microblog Posts on Attitude Towards Vaccination*, we build and benchmark classical text classifiers (Naive Bayes, Logistic Regression, linear SVM) under different Chinese representations (jieba words vs. character n-grams) on Weibo posts. The supplied corpus has **seven emotion classes** (IDs 0–6) rather than the paper's binary vaccine polarity, so this is a 7-class emotion study. Our central contribution is a **rigorous, leakage-free evaluation**: we show that the raw file's evaluation is inflated by duplicated posts, fix it, and report honest metrics with confidence intervals.

2 Dataset

Corpus Mixed-topic Chinese Weibo corpus, **21,173 posts**, columns *review, label*. Only ≈5% of posts contain COVID-related words, and there is no date column, so it is **not** a vaccination-specific dataset and no time-trend analysis is possible.
Cleaning 10,154 exact-duplicate texts (**48%** of rows; one post appears 43 times); after normalised matching and removal of 86 texts with contradictory labels → **10,565 unique posts** (50.1% removed). Class 4 shrinks from 3,072 to 387, so class weights and macro-F1 are used.
Split Stratified, seed 42: train 6,339 · validation 2,113 · test 2,113, with zero text overlap (checked by unit test).
Label names The dataset provides only IDs. Names (fear, disgust, happiness, surprise, like/gratitude, sadness, anger) are **tentative**, inferred from top-weighted n-grams (e.g. 怕/吓→0, 恶心→1, 哈/笑→2, 惊/居然→3, 感谢/致敬→4, 泪/悲→5, 怒/骂→6).

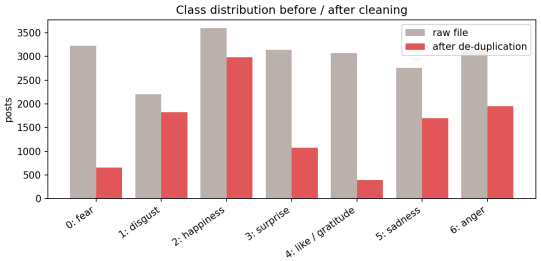


Figure 1. Class distribution before/after de-duplication.

3 Approach

Preprocessing NFKC normalisation; URLs and @mentions removed; repeated punctuation/characters collapsed; Weibo emoticons such as [good], [委屈] are **kept** as tokens (EMO_x) – the original pipeline deleted them (ablation: +3.3 pts accuracy). **Features** TF-IDF (sublinear) on jieba words (1–2-grams), on character 1–3-grams, or both. **Models** Complement Naive Bayes; Logistic Regression (C∈{1,3,10}); Linear SVM (C∈{0.1,0.3,1}); all with balanced class weights → 19 candidates + majority baseline. **Protocol** candidates fitted on train, ranked by validation macro-F1; winner re-fitted with sigmoid-calibrated probabilities and scored **once** on test; 95% bootstrap CIs (1,000 resamples) and 5-fold CV on train+val.

Implementation Python 3.11, scikit-learn, jieba, pandas, matplotlib, Streamlit; seed 42. Modules in `src/`: preprocess, data, models, train, evaluate, leakage_audit, predict; `app.py` web demo; 11 pytest tests; `python run_all.py` reproduces all results.

4 Validation Results

Classifier (best setting)	Features	Val Acc	Val macro-F1
Linear SVM (C=0.3)	char 1–3-gram	0.699	0.695
Logistic Regression (C=3)	char 1–3-gram	0.690	0.688
Linear SVM (C=0.1)	word + char	0.689	0.685
Linear SVM (C=0.3)	jieba words	0.647	0.638
Naive Bayes	jieba words	0.634	0.610
Majority-class baseline	–	0.283	0.063

Table 1. Best configurations on the validation set (full grid: results/model_comparison.csv).

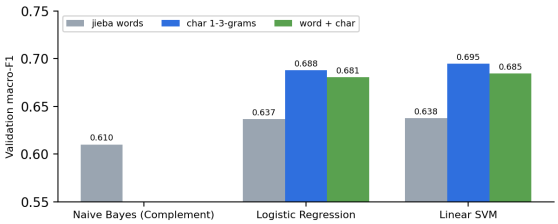


Figure 2. Best validation macro-F1 per model family and feature set.

Observations. Character n-gram features lift every linear model by 5–6 points over jieba words; combining word and char features did not help further. Differences of about 1 point between top candidates are within validation noise (2,113 posts), so we pick by macro-F1 but do not over-interpret the ranking. **Demo.** `streamlit run app.py` opens a web app: pick an example or type a post to see the predicted class with calibrated probabilities for all seven classes, upload a CSV for batch prediction, and view the honest-results dashboard; a CLI (`python -m src.predict_demo`) is also provided.

5 Test Set Evaluation

Selected strictly on validation macro-F1: **Linear SVM, char 1–3-grams, C=0.3**. Evaluated once on the held-out test set (2,113 unique posts).

Metric	Value
Accuracy	0.687 (95% CI 0.668–0.706)
Macro-F1	0.672 (CI 0.650–0.693)
5-fold CV macro-F1	0.683 ± 0.016
Majority baseline acc.	0.283
Best / worst class F1	like 0.86 / surprise 0.51
COVID-keyword subset (n=127)	acc 0.795 (indicative only)

Table 2. Test-set summary. The keyword subset is small and skewed to class 4.



Figure 4. Per-class F1 on test set.

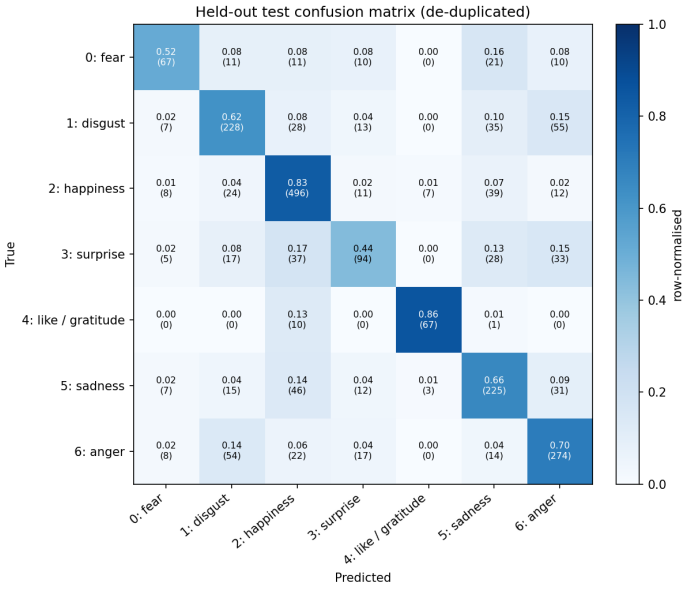


Figure 3. Row-normalised test confusion matrix.

6 Key Finding: Duplicate Leakage Inflated the Original Evaluation

Because 48% of rows are repeated posts, a random split on the raw file puts the same post in train and test (**56.7%** of test rows also in train). We re-ran both an original-style model (word TF-IDF + LR, emoticons deleted) and our improved model under both protocols (`src/leakage_audit.py`):

Model	Original split	Clean split
Original-style (word LR)	78.5%	58.9%
Improved (char SVM)	84.9%	**68.7%**

On the original split the original-style model scored 91.9% on leaked rows but only 60.9% on unseen rows. The improved model is ≈10 accuracy points better under honest evaluation.

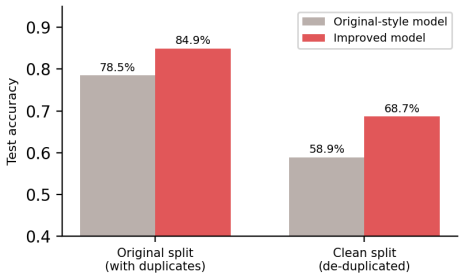


Figure 5. Test accuracy by protocol.

7 Error Analysis

The most frequent confusions are **disgust↔anger** (55 and 54 test posts) and **sadness→happiness** (46): neighbouring emotions share negative vocabulary, and sadness/happiness posts often carry mixed cues (laughter markers around a sad statement). **Surprise** has the lowest recall (0.44): it has few distinctive words and is spread over happiness, anger and sadness. Among the 100 most confident errors, many are very short posts (mean length 24 characters), e.g. 加油，武汉加油 (true: gratitude, predicted: happiness) or 我还以为是别墅呢。 (true: disgust, predicted: anger) – sarcasm and context-dependent emotion that bag-of-n-gram models cannot resolve (`results/confident_errors.csv`).

8 Scope, Conclusions & Limitations

Conclusions. (1) Character n-grams beat jieba word features by ≈6 points (0.695 vs 0.638 macro-F1): they avoid segmentation errors on slang, typos and emoticons. (2) Keeping emoticons helps (+3.3 pts). (3) Linear SVM and LR are within noise of each other; both far exceed Naive Bayes. (4) The biggest accuracy gain came from *fixing the evaluation*, not the model. **Scope.** No LSTM and no monthly trend analysis were done: the dataset has no dates or vaccine filter, and no pre-trained models could be downloaded; we do not claim the reference paper's trend results. **Limitations.** Tentative class names; mixed-topic corpus (results do not transfer to vaccination posts); single split seed; ≈69% is near the ceiling for n-gram models. **Future work.** Chinese pre-trained transformers (e.g. RoBERTa-wwm), a vaccine-specific dated dataset, and a neutral class.

9 Team Contributions

Harsha R (PES1UG24AM115) – text cleaning and emoticon handling, de-duplication and splitting, classical models and hyper-parameter search, Streamlit/CLI demo.

Darisi Anantha Venkata Sreeram (PES1UG24AM076) – evaluation harness (bootstrap CIs, CV, ablation), leakage audit, error analysis, unit tests, documentation.